Abdoali Mazhar Zakir

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PROFESSIONAL SUMMARY

Final-year Software Engineering (Honours) student at UNSW who specialises in building scalable data systems and applied machine learning solutions. I've shipped production applications across the stack, including financial knowledge graphs and AI-driven platforms, while prioritising clean architecture and automation. Recently, I streamlined deployment pipelines to cut manual workflows by 30% and significantly improve system reliability. Currently, I'm at UNSW developing machine learning models for complex Citation Link prediction.

EDUCATION

Bachelor of Engineering (Honours) in Software Engineering — **September 2024 – December 2026**
University of New South Wales

Bachelor of Science in Computer Science & Engineering — **July 2022 – May 2024**
International Centre for Applied Sciences (ICAS),

RELEVANT EXPERIENCE

Project Engineer Intern
Tata Consultancy Services (TCS) — **January 2026 – February 2026**

- Built a Financial Crime Compliance knowledge graph by transforming large-scale transactions and account datasets into structured graph entities using Neo4j, enabling advanced network analysis for fraud detection.
- Collaborated within a multi-disciplinary engineering team following Agile delivery practices, contributing to project planning, milestone reviews, and final technical presentations to industry stakeholders
- Applied graph similarity algorithms to identify hidden relationships between accounts, supporting fraud-risk analysis across complex financial networks.
- Presented data findings and methodology to industry stakeholders, translating technical graph metrics into plain-language risk insights within an Agile delivery team.

Data Analytics Intern
Futures Sport & Entertainment (IPG DXTRA / Omnicom) — **July 2026 – September 2026**

- Joined the Data Analytics team at Futures Sport & Entertainment, supporting data-driven sponsorship and audience insight work across sport and entertainment clients.
- Working across data collection, cleaning, and analysis to support client-facing reporting and insight generation, three days per week through the internship period.

Data and Automation Developer
Enigma Technical Society — **September 2023 – May 2024**

- Built internal reporting and automation scripts in Python to track and analyse society operations data, cutting manual reporting effort by 30%.
- Designed and maintained a PostgreSQL data layer for event and membership data, ensuring clean, query-ready datasets for a 6-member Agile team.
- Automated data-refresh and deployment workflows using GitHub Actions CI/CD pipelines, reducing manual errors in recurring data updates.
- Contributed to brainstorming sessions and technical documentation, reinforcing code quality and collaborative development practices.

PROJECTS

Sentinel ETL — Self-Healing Data Platform **June 2026**
Personal Project

- Designed and built an enterprise-inspired ETL platform implementing Medallion Architecture (Bronze/Silver/Gold layers) to automatically ingest, validate, repair, and transform datasets.
- Built a self-healing data quality engine (Python/FastAPI) that automatically detects and remediates missing values, duplicates, schema drift, and type mismatches, with full audit logging and data lineage tracking for every transformation.
- Developed a Next.js/TypeScript/Tailwind analytics dashboard to visualise data quality scores, pipeline health, and dataset history.
- Used DuckDB, Pandas, and PyArrow for the data processing layer, PostgreSQL for metadata storage, and Docker with GitHub Actions for CI/CD automation.

GitHub: https://github.com/AbdoZak786/Sentinal_ETL

TerraCast – Viticulture Weather & Advisory Platform **February 2026 – May 2026**
UNSW

- Built serverless ETL pipelines (AWS Amplify/Lambda) to ingest, clean, and structure multi-source weather data (rainfall, frost, temperature, wind) for downstream forecasting.
- Developed responsive frontend features using Next.js (App Router), including dashboards and recommendation views, with state management and API integration.
- Implemented server-side generative AI features using OpenAI API (gpt-4o-mini), including a viticulture assistant and forecast-based recommendation system using structured outputs.
- Collaborated in an Agile team, contributing to API design, system integration, and technical documentation, including testing strategy and deployment planning.

Thesis — Citation Prediction using CDCHGN Knowledge Graphs **2026**

- Designing heterogeneous graph neural network models to predict citation links using community detection and node embedding techniques — large-scale graph data processing directly applicable to MongoDB's Atlas Graph and aggregation pipeline internals.

CORE COMPETENCIES

Languages: **Python**, JavaScript, C++, SQL, Cypher, HTML/CSS.

Frameworks & Libraries: Node.js, React, Jest, Pytest, PyTorch

Databases: PostgreSQL, MySQL, Neo4j(graph)

Tools & Platforms: Docker, AWS (Terraform, S3), Postman, Neo4j

VOLUNTEERING AND EXTRACURRICULAR

Lead Volunteer **June 2023- May 2024**
Voluntary Service Overseas (VSO), Manipal

Secretary **March 2024- Present**
Social Football Society, UNSW

Sports Captain **June 2023- May 2024**
International Centre for Applied Science, Manipal University